



EXPERIMENT 1.5

Accelerating masses

Aim

To determine the relationship between mass and acceleration.

Materials

Dynamics trolley

String

Mass hanger and brass 50 g masses

Several 1 kg masses

Desk-mountable pulley wheel with clamp

Electronic balance

Motion sensor or stopwatch

Tape measure or ticker timer

Power supply

Ticker tape

Method

- 1 Clamp the pulley wheel to the edge of the desk. Try to arrange the largest height above the floor as possible.
- 2 Attach one end of the string to the dynamics trolley and the other end to the mass hanger, carrying a total of approximately 200 g of mass.
- 3 Hang the masses over the pulley so they can pull the trolley along as they fall to the floor.
- 4 Record the motion of the trolley as the masses fall using either a motion sensor, timing with a stopwatch or recording the motion on ticker tape.
- 5 Successively add 1 kg masses to the trolley and repeat your measurements several times.

Results

- 1 Determine the acceleration of the trolley for your method.
- 2 If you used a motion sensor, use software (see Experiment 1.2) to determine the acceleration directly or from the gradient of a velocity–time graph.
- 3 If you used a stopwatch, calculate the acceleration as $(2 \times \text{the distance travelled} \div \text{time squared})$.
- 4 If you used a ticker timer, use the ‘every fifth dot method’ as per Experiment 1.1 to divide the tape into sections. Determine the speed of each section by dividing the distance covered by 0.1. Plot a speed–time graph and determine the acceleration from the gradient of the graph.
- 5 Plot a graph of acceleration versus total mass. This should give a truncated, or inverse, graph.
- 6 Plot a second graph of acceleration versus the inverse of mass. This should give a straight-line graph.

Discussion

Analyse and evaluate your results.

Conclusion

What is the relationship (or formula) between mass and acceleration?

